

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Implement the following Servlet applications with Cookies and Sessions.

Practical No. IV

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Class: TYIT	Batch:
Date of Assignment: 18/07/2026	Date/Time of Submission: 22/07/2026

2(A)

Aim: Develop a simple JSP application to display values obtained from the use of intrinsic objects of various types.

Code:

index.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<html><head><title> JSP Page</title></head>
<body>
    <h1>Use of Intrinsic Objects in JSP</h1>
    <h1>Request Object </h1>
    Query String <%=request.getQueryString() %><br>
    Context Path <%=request.getContextPath() %><br>
    Remote Host <%=request.getRemoteHost() %><br>
    <h1>Response Object </h1>
    Character Encoding Type <%=response.getCharacterEncoding() %><br>
    Content Type
    <%=response.getContentType() %><br>
    Locale <%=response.getLocale() %><br>
    <h1>Session Object </h1>
    ID <%=session.getId() %><br>
    Creation Time <%=new java.util.Date(session.getCreationTime()) %><br>
    Last Access Time<%=new java.util.Date(session.getLastAccessedTime()) %><br>
</body>
</html>
```

Output:

Use of Intrinsic Objects in JSP

Request Object

Query String null
Context Path /pract4
Remote Host 0:0:0:0:0:0:1

Response Object

Character Encoding Type UTF-8
Content Type text/html;charset=UTF-8
Locale en_IN

Session Object

ID 43d65f9e5f255c8a5cda5c68a947
Creation Time Sat Jul 18 13:30:08 IST 2026
Last Access Time Sat Jul 18 13:30:08 IST 2026

2(B)

Aim: Develop a simple JSP application to pass values from one page to another with validations. (Name-txt, age-txt, hobbies-checkbox, email-txt, gender-radio button).

Code:

index.jsp

```
<!DOCTYPE html>
<html>
  <head>
    <meta https-equiv="Content-Type" content="text/html;charset=UTF-8">
    <title>JSP Registration</title>
  </head>
  <body>
    <h1>Registration Form</h1>
    <form action="val.jsp" method="post">
      <table style="width:50%">
        <tr>
          <td>Full Name</td>
          <td><input type="text" name="fullname"/></td>
```

```

        </tr><tr>
            <td>Age</td>
            <td><input type="text" name="age"/></td>
        </tr><tr>
            <td>E-Mail</td>
            <td><input type="text" name="email" size="20"/></td>
        </tr><tr>
            <td>Gender</td>
            <td><input type="radio" name="gender" value="Male">Male
                <input type="radio" name="gender" value="Female">Female
            </td>
        </tr><tr>
            <td>Hobbies</td>
            <td><input type="checkbox" name="hb" value="Acting"/>Acting
                <input type="checkbox" name="hb" value="Dancing"/>Dancing
                <input type="checkbox" name="hb" value="Singing"/>Singing
                <input type="checkbox" name="hb" value="Drawing"/>Drawing
            </td>
        </tr>
    </table>
    <input type="submit" value="Register"/>
</form>
</body>
</html>
val.jsp
<%@page language="java" contentType="text/html" pageEncoding="UTF-8" %>
<%!
    int ageInNumbers;
    private static final String EMAIL_REGEX = "^([\\w-\\+]+)(\\.([\\w-\\+]+)*@[\\w-\\+](\\.([\\w-\\+]+)*)(\\.[a-z]{2,})$";
%>
<%
    String name = request.getParameter("fullname");
    String age = request.getParameter("age");
    String email = request.getParameter("email");
    String gender = request.getParameter("gender");
    String hb[] = request.getParameterValues("hb");
    if (name == null || name.isEmpty() || age == null || age.isEmpty() || email == null ||
        email.isEmpty() || gender

```

```

        == null || gender.isEmpty()) {
    out.println("<font color=red>Please fill all the fields</font><br>");
}
if (!email.matches(EMAIL_REGEX)) {
    out.println("<font color=red>Correct Your Email Address</font><br>");
}
try {
    ageInNumbers = Integer.parseInt(age.trim());
} catch (NumberFormatException e) {
    out.println("<font color=red>Age must be numbers</font><br>");
}
if (ageInNumbers < 18 || ageInNumbers > 60) {
    out.println("<font color=red>Age must be between 18 and 60</font><br>");
}
%>
Your Extended Information is as follows:<br><br>
Full Name<b>:<%=name%></b><br>
Age<b>:<%=age%></b><br>
EMail<b>:<%=email%></b><br>
Gender<b>:<%=gender%></b><br>
Hobbies<b>:
<%
    if (hb != null && hb.length != 0) {
        for (int i = 0; i < hb.length; i++) {
            out.println(hb[i]);
        }
    }
%>

```

web.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

    xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee
http://xmlns.jcp.org/xml/ns/javaee/web-
app_3_1.xsd"

    version="3.1">
    <welcome-file-list>
        <welcome-file>index.jsp</welcome-file>
    </welcome-file-list>
</web-app>

```

Output:

Registration Form

Full Name	divyanjali
Age	19
E-Mail	divs@gmail.com
Gender	☐ Male ☒ Female
Hobbies	☐ Acting ☐ Dancing ☐ Singing ☒ Drawing
Register	

Your Extended Information is as follows:

Full Name:**divyanjali**
 Age:**19**
 EMail:**divs@gmail.com**
 Gender:**Female**
 Hobbies: **Drawing**

2(C)

Aim: Create a registration and login JSP application to register and authenticate the user based on username and password using JDBC.

Code:

index.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
  <title>Registration</title>
</head>
<body>
<center>
<h2>User Registration</h2>
<form action="registerprocess.jsp" method="post">
Username :
<input type="text" name="username" required>

<br><br>
Password :
<input type="password" name="password" required>
<br><br>
<input type="submit" value="Register">
```

```

<input type="reset" value="Clear">
</form>
<br>
<a href="login.jsp">Already Registered? Login</a>
</center>
</body>
</html>

```

login.jsp

```

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
    <title>Registration</title>
</head>
<body>
<center>
<h2>User Registration</h2>
<form action="registerprocess.jsp" method="post">
Username :
<input type="text" name="username" required>
<br><br>
Password :
<input type="password" name="password" required>
<br><br>
<input type="submit" value="Register">
<input type="reset" value="Clear">
</form>
<br>
<a href="login.jsp">Already Registered? Login</a>
</center>
</body>
</html>

```

loginprocess.jsp

```

<%@ page import="java.sql.*" %>
<%@ page import="database.DBConnection" %>
<%
String username = request.getParameter("username");
String password = request.getParameter("password");
Connection con = DBConnection.getConnection();
PreparedStatement ps = con.prepareStatement(

```

```

"SELECT * FROM users WHERE username=? AND password=?");
ps.setString(1, username);
ps.setString(2, password);
ResultSet rs = ps.executeQuery();
if(rs.next()){
    session.setAttribute("username", username);
    rs.close();
    ps.close();
    con.close();
    response.sendRedirect("welcome.jsp");
}else{
    rs.close();
    ps.close();
    con.close();
    out.println("<h2>Invalid Username or Password</h2>");
}
%>

```

logout.jsp

```

<%
session.invalidate();
response.sendRedirect("login.jsp");
%>

```

register.jsp

```

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
    <title>Registration</title>
</head>
<body>
<center>
<h2>User Registration</h2>
<form action="registerprocess.jsp" method="post">
Username :
<input type="text" name="username" required>
<br><br>
Password :
<input type="password" name="password" required>
<br><br>
<input type="submit" value="Register">

```

```

<input type="reset" value="Clear">
</form>
<br>
<a href="login.jsp">Already Registered? Login</a>
</center>
</body>
</html>

```

registerprocess.jsp

```

<%@ page import="java.sql.*" %>
<%@ page import="database.DBConnection" %>
<%
String username = request.getParameter("username");
String password = request.getParameter("password");
Connection con = DBConnection.getConnection();
PreparedStatement ps = con.prepareStatement(
"INSERT INTO users(username,password) VALUES(?,?)");
ps.setString(1, username);
ps.setString(2, password);
int i = ps.executeUpdate();
ps.close();
con.close();
if(i>0){
    response.sendRedirect("login.jsp");
}else{
    out.println("Registration Failed");
}
%>

```

welcome.jsp

```

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%
String user=(String)session.getAttribute("username");
if(user==null)
{
    response.sendRedirect("login.jsp");
}
%>
<!DOCTYPE html>
<html>
<head>
    <title>Welcome</title>

```

```

</head>
<body>
<center>
<h1>Welcome <%=user%></h1>
<br><br>
<h3>Login Successful</h3>
<br>
<a href="logout.jsp">Logout</a>
</center>
</body>
</html>

```

DBConnection.java

```

package database;
import java.sql.Connection;
import java.sql.DriverManager;
public class DBConnection {
    public static Connection getConnection() {
        Connection con = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/jspdb",
                "root",
                "Root@2007");

            System.out.println("Database Connected Successfully");
        } catch (Exception e) {
            e.printStackTrace();
        }
        return con;
    }
}

```

User Registration

Registration and Login System

Username :

Password :

[Register Here](#)

[Login Here](#)

Register

Clear

[Already Registered? Login](#)

User Login

Username :

Password :

[New User? Register](#)

Welcome divs

Login Successful

[Logout](#)